

Roll No. :

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B. Tech CSE**

Semester: V

Name of the Course: Security and Auditing

Course Code: TCS 595

Time: 90 Minutes

Maximum Marks: 50

Note:

- i. Answer all the questions by choosing any one of the sub questions.
- ii. Each questions carries 10 marks.

Q1	(10 Marks)	
(a)	Define a security hole. How it can be fixed through the releasing of a security patch?	CO1
	OR	
(b)	Define social engineering attacks. Discuss three common techniques used by attackers.	CO2
Q2	(10 Marks)	
(a)	Explain the concept of White Hat Search Engine Optimization (SEO) and how attackers abuse SEO.	CO2
	OR	
(b)	Discuss five common hacker's tools and their usage in cyber attacks.	CO4
Q3	(10 Marks)	
(a)	Write short notes on IT Act 2000 Provisions relevant to cybersecurity in India	CO2
	OR	
(b)	Differentiate between Copyright Law, Patent Law, and Intellectual Property Rights (IPR) in software protection.	CO3
Q4	(10 Marks)	
(a)	A healthcare organization needs to comply with HIPAA. How can they map HIPAA controls to the NIST Cybersecurity Framework?	CO6
	OR	
(b)	Analyze how HIPAA violations can occur through phishing or ransomware in healthcare. Provide an example case.	CO5
Q5	(10 Marks)	
(a)	Differentiate between the following types of firewalls: Packet-filtering, stateful, proxy, and application-level.	CO3
	OR	
(b)	Compare phishing, pharming, and smishing in terms of techniques, impact, and mitigation strategies.	CO5

